



Fermented corn bran: A by-product with improved total phenolic content and antioxidant activity

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ABSTRACT

Corn bran is a milling by-product of the corn industry; however, it has been considered a potential source of phenolic compounds. This study was conducted to evaluate the potential of solid-state fermentation with *Lactobacillus reuteri* and *Lactobacillus plantarum* for improving the phenolic compounds and antioxidant activity of corn bran. The fermentation caused a significant increase in the total phenolic content (TPC) of fermented corn brans compared to the control sample. Highest TPC was obtained after 48 h of fermentation. The antioxidant activity, measured through different methodologies (DPPH, ABTS, ORAC assays), was higher in the fermented corn bran samples than in the control one. Fermentation increased the concentration of free phenolic acids, especially ferulic acid and *p*-coumaric acid, in the fermented samples. Fourier-transform infrared spectroscopy indicated that fermentation promoted the degradation of some ester bonds in the substrate structure, which could lead to the release of free phenolic acids. Meanwhile, scanning electron microscopy showed that the fermentation led to the formation of a loose surface structure in the fermented samples. The results revealed that solid-state fermentation was an effective method for the enhancement of phenolic compounds, suggesting that this bioprocessing technology may help improve the antioxidant activity of fermented corn bran.

1. Introduction

The worldwide production of corn bran, driven by the corn processing industries such as starch and ethanol production, has reached 8.87 billion tons in 2021 reported by Expert Market Research (EMR Inc., 2022). Corn bran is an abundant by-product of the corn industries, mostly consisting of the outer layers of the corn kernel (pericarp, testa, and aleurone) along with 10–25% of impacted starch. It is primarily used as a low-value ingredient in livestock feed; however, it has been considered a great source of phytochemicals such as phytosterols, dietary fiber, carotenoid pigments, and phenolic compounds (Acosta-Estrada, Gutiérrez-Urbe, & Serna-Saldívar, 2014; Saeed et al., 2021). Recently, phenolic compounds have received great attention because of their physiological functional properties including antioxidant, antimicrobial, anti-inflammatory, and anti-cancer activities (Tungmunthum, Thongboonyou, Pholboon, & Yangsabai, 2018).

Although corn bran has the highest content of phenolic compounds in comparison with other cereal brans (Galanakis, 2018), most of its

phenolic compounds are bound and insoluble, as about 85% of them are bound to cell wall structural compounds such as cellulose, hemicellulose, lignin, pectin, and rod-shaped structural proteins. Phenolic acids are linked to lignin by ether bonds (via their hydroxyl groups) and to carbohydrates and structural proteins by ester bonds (via their carboxyl groups) (Acosta-Estrada et al., 2014). These interactions might have an adverse impact on polyphenols bioavailability *in vivo*. Nevertheless, some research suggested that solid-state fermentation by yeasts or lactic acid bacteria (LAB) could significantly increase the content and bioavailability of bioactive compounds (Liu, Chen, Shao, Wang, & Zhan, 2017). During the solid-state fermentation, microorganisms have been shown to release phenolic compounds directly by producing enzymes such as esterases, amylases, beta-glucosidases, and xylanases which are able to cleave the bonds between phenolic compounds and the plant cell wall components (Oliveira, Cipolatti, Furlong, & Soares, 2012; Verdusco-Oliva & Gutierrez-Urbe, 2020).

The potential of solid-state fermentation has been investigated for the improvement of the total phenolic content and antioxidant activity

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of brown rice (Li et al., 2019), barley and oat (Hole et al., 2012), quinoa (Ayyash et al., 2019), lupin (Bartkiene, Krungleviciute, Juodeikiene, Vidmantienė, & Maknickienė, 2015), rice bran (Nur, Dang, Anisah, Shaiful, & Long, 2015), oat (Wu et al., 2018), wheat (Ripari, Bai, & Gänzle, 2019), wheat bran (Spaggiari et al., 2020), and sorghum (Svensson, Sekwati-Monang, Lutz, Schieber, & Ganzle, 2010) using different LAB. In the available literature, there are few references regarding strategies for the release and extraction of polyphenolic compounds from corn bran. The purpose of the present study is to evaluate the ability of selected LAB to improve the phenolic content and antioxidant activity of corn bran by solid-state fermentation. In addition, it is aimed at investigating the influence of the fermentation on the morphological structure and changes of the substrate functional groups using scanning electron microscopy (SEM) and Fourier-transform infrared spectroscopy (FT-IR), respectively.

2. Materials and methods

2.1. Materials

Corn bran (protein, 7 g/100 g; fat, 4 g/100 g; starch, 10 g/100 g) was kindly supplied by Zar Industrial and Research Group (Hashtgerd, Iran), which was used as a substrate for the solid-state fermentation. The dry sample was ground to a fine powder using a high-speed grinder, packaged in polyethylene zipper bags, and stored at -20°C to be preserved prior to use for fermentation.

Phenolic acid standards (ferulic acid, *p*-coumaric acid, caffeic acid, vanillic acid, and syringic acid), 2,2-diphenyl-1-picrylhydrazyl (DPPH), 2,2'-azino-bis (3-ethylbenzothiazoline-6-sulfonic acid) (ABTS), fluorescein sodium, 2,2'-azobis(2-methyl-propionamide) dihydrochloride (AAPH), 6-hydroxy-2,5,7,8-tetramethylchroman-2-carboxylic acid (Trolox), gallic acid, Folin-Ciocalteu reagent, sodium carbonate, all, of analytical- or HPLC-grade, were purchased from Sigma-Aldrich (Steinheim, Germany).

2.2. Starter microorganisms

Lactobacillus plantarum (DSM 20179) and *Lactobacillus reuteri* (PTCC 1655, Persian Type Culture Collection, Tehran, Iran) used as fermentative strains, were cultured in MRS broth (Merck, Darmstadt, Germany) at 37°C for 24 h.

2.3. Solid-state fermentation conditions

20 g of the corn bran was washed and soaked in distilled water (moisture content was 70 g/100 g) in a 500-mL Erlenmeyer flask and subsequently sterilized (121°C , 15 min). After cooling at ambient temperature ($21\text{--}23^{\circ}\text{C}$), the sterilized substrate was inoculated with 2 mL of the LAB suspension containing 10^7 CFU mL^{-1} . After mixing, fermentation was carried out at 35°C for 72 h. To accelerate the release of the fermentation heat, the medium was stirred at 24-h intervals during the fermentation (Ayyash et al., 2019).

2.4. Extraction of free phenolic compounds

The fermented corn bran samples were taken out of the Erlenmeyer flasks at 12-h intervals and dried at 60°C for 24 h using a vacuum oven (Bangar et al., 2022). Phenolic compounds were extracted from the dried substrates. The optimal conditions of the extraction process were chosen according to previous optimization studies. The dried fermented samples (2 g) were extracted with 50 mL of ethanol-water (70:30, v/v) at 45°C for 1 h (Punia et al., 2020). After centrifugation at $1700\times g$ (Centrifuge 5810 R, Eppendorf, Germany) for 10 min, the supernatants were filtered, and the filtrates were collected as the phenolic solution (Chen et al., 2020).

2.5. Determination of total phenolic content (TPC)

The Folin-Ciocalteu phenol reagent was used to estimate the TPC of each extract according to Liu et al. (2017) with slight modification. 0.2 mL of each extract was mixed with 1 mL of the diluted Folin-Ciocalteu reagent. After 5 min, 2 mL of sodium carbonate 20 g/100 mL was added. The mixture was stirred and allowed to stand at ambient temperature in a dark place for 60 min. Then, the absorbance value was recorded at 765 nm using a UV-Vis spectrophotometer (Lambda 365, PerkinElmer, the USA). The analyses were done in triplicate. The results were quantified using external calibration expressed as mg of gallic acid equivalents per g of sample on a dry weight basis (mg GAE/g dwb).

2.6. Antioxidant activity

2.6.1. Radical scavenging by DPPH assay

The antioxidant activity of the polyphenol extracts was examined by DPPH free radical scavenging assay as described by Anticona, Lopez-Malo, Frigola, Esteve, and Blesa (2021). Briefly, 50 μL was taken from the sample extract, and a mixture was formed in glass test tubes by adding 1450 μL of DPPH (60 $\mu\text{mol/L}$). The mixture was shaken vigorously and incubated at ambient temperature in a dark place for 30 min. Then, the absorbance value was determined at 515 nm. Trolox was also used as the standard (50–600 $\mu\text{mol/L}$). The procedure was triplicated, and the scavenging activity percentage of the samples was calculated against the blank (which contained all the reagents without the polyphenol extract) by the following formula:

$$\text{DPPH}^{\bullet} \text{ scavenging activity (\%)} = [(A_{517} \text{ blank} - A_{517} \text{ Extract}) / A_{517} \text{ blank}] \times 100$$

Finally, the results were expressed as $\mu\text{mol/L}$ Trolox equivalents ($\mu\text{mol TE/g dwb}$).

2.6.2. radical scavenging by ABTS assay

ABTS assay was used for the assessment of the radical scavenging activity of the antioxidant extracts during fermentation, according to the method described by Ayyash et al. (2019) with slight modification. A stock solution of $\text{ABTS}^{\bullet+}$ was prepared by oxidizing 25 mL ABTS (7 mmol/L) with 440 μL potassium persulfate $\text{K}_2\text{S}_2\text{O}_8$ (140 mmol/L). The solution was diluted with ethanol until an absorbance of 0.70 ± 0.02 was reached at 734 nm after equilibration at 30°C . 30 μL of the antioxidant extract was added to 3 mL of $\text{ABTS}^{\bullet+}$ and incubated at 30°C . After 10 min of the standing in a dark place, the absorbance value was read at 734 nm. 30 μL of distilled water was used as blank. $\text{ABTS}^{\bullet+}$ scavenging activity (%) was calculated using the equation below:

$$\text{ABTS}^{\bullet+} \text{ scavenging activity (\%)} = [(A_{734} \text{ Blank} - A_{734} \text{ Extract}) / A_{734} \text{ Blank}] \times 100$$

2.7. Oxygen radical absorbance capacity (ORAC)

The ORAC procedure was carried out on a spectrofluorometer (Victor X3, PerkinElmer, the USA) with 96-well plates (Sero-Wel, Bibby Sterilin Ltd., Stone, UK), which was equipped with an incubator. The reaction was conducted in sodium phosphate buffer 75 mmol/L (pH 7.0). 50 μL of the diluted antioxidant extract (1:250) and 50 μL of fluorescein sodium ($\text{C}_{20}\text{H}_{10}\text{Na}_2\text{O}_5$) 78 nmol/L were first pipetted into each well and incubated at 37°C for 15 min. The phosphate buffer was used as a blank, and Trolox solution (20 $\mu\text{mol/L}$) was also used as standard. 25 μL of AAPH 221.25 mmol/L was then added as the free radical generator. The plate reader was programmed to read fluorescence every 5 min ($\lambda_{\text{excitation}} = 485 \text{ nm}$, $\lambda_{\text{emission}} = 538 \text{ nm}$) up to 18 cycles. The fluorescence reading for each cycle was saved to calculate the areas under the fluorescence decay curve (AUC) by the following equation:

$$\text{AUC} = (0.5 + f_1 / f_1 + f_2 / f_1 + f_3 / f_1 + \dots + f_n / f_1) \times 5$$

where f_1 is the initial fluorescence reading at time zero, f_n denotes the fluorescence reading at cycle n , and 5 shows the cycle duration (min) (Zulueta, Esteve, & Frígola, 2009).

2.8. HPLC-UV system for the analysis of phenolic acids

Phenolic acids were detected and quantified by an HPLC system (Agilent Technologies 1120 Compact LC) integrated with a UV detector equipped with a C18 column (250 × 4.6 mm, particle size 5 µm; Luna PFP2, Phenomenex) which was controlled at 25 °C. The mobile phase consisted of water containing 0.1% v/v formic acid (solvent A) and acetonitrile:solvent A (60:40 v/v) mixture (solvent B) with a flow rate of 0.8 mL/min. A gradient elution was applied as follows: 0–10 min, linear 100 to 85% solvent A; 10–20 min, linear from 85 to 82% solvent A; 20–50 min, linear from 82 to 0% solvent A; 50–70 min, isocratic 0% solvent A; 70–75 min, linear from 0 to 100% solvent A; 75–80 min, isocratic 100% solvent A.

Specific phenolic acids were detected by comparing their retention times with those of the external standards (ferulic, *p*-coumaric, caffeic, vanillic, and syringic acids) at 275 nm. They were also quantified using the calibration curve obtained from the retention times and UV absorbance values of the external standards at 275 nm. The injection volume was 20 µL, and the antioxidant extracts were analyzed in duplicate.

Prior to the HPLC-UV analysis, a solid phase extraction (SPE) clean-up process was used on C18 cartridges (55 µm, 70 Å, Phenomenex, Steinheim, Germany). Briefly, a 5 mL antioxidant extract was diluted to 20 mL by Mili-Q containing 1% formic acid. C₁₈ cartridges were pre-conditioned by loading 2 × 3 mL of methanol and 2 × 3 mL of Milli-Q water. After 2 min, the diluted antioxidant extracts were loaded. Then cartridges were washed with 5 mL Mili-Q containing 1% formic acid. Polyphenols fractions were recovered using 3 mL of 0.1% formic acid in methanol. All methanolic fractions were dried under a light stream of nitrogen, redissolved in 1 mL of 0.6 mol/L HCL in methanol (75% v/v), and were filtered through a 0.22 µm nylon membrane for subsequent analysis on the HPLC-UV system (Anticona, Lopez-Malo, Frígola, Esteve, & Blesa, 2022).

2.9. FT-IR analysis

An Avatar spectrometer (Thermo Nicolet, the USA) was applied to evaluate the corn bran samples both before and after solid-state fermentation. The samples were scanned in the wavenumber range of 4000–650 cm⁻¹ transmittance mode at a resolution of 4 cm⁻¹ (Silva et al., 2021).

2.10. Scanning electron microscopy (SEM)

A MIRA 3 scanning electron microscope (Tescan, Czech Republic) at 15 kV acceleration voltage was used to analyze the surface morphology of the fermented substrates and the control. 0.2 g of each sample was taken, fixed with glutaraldehyde 2.5 g/100 mL, and dehydrated at a gradient of ethanol (30, 50, 70, 80, 90, and 100%). The samples were placed on copper support and coated with a conductive layer of gold powder using a cathodic ion spray (de Olmos, Garro, & Garro, 2022).

2.11. Data analysis

All measurements were done in triplicate, and the results were reported as mean values ± standard deviation (SD). Data were analyzed by one-way analysis of variance (ANOVA) using SPSS version 21 (IBM, Chicago, the USA) while comparison between means was achieved using Duncan's multiple range method. Statistical differences were considered, and the coefficients of determination (R^2) were calculated to determine the relationship between the two variables. Differences were considered significant when $p < 0.05$.

3. Results and discussion

3.1. TPC

The TPCs of the fermented and control corn bran samples are presented in Table 1. The Folin-Ciocalteu method is widely used to examine the TPC of antioxidant extracts in vitro. The TPC of the control corn bran was found to be 0.63 mg GAE/g dry matter, conforming to that previously reported by Hussain et al. (2021).

The samples fermented by the LAB had higher TPCs than the control. A significant increase in TPC during fermentation is an important indicator of the positive effect of the solid-state fermentation on the substrate (Punia et al., 2020). In unfermented corn bran, a majority of ferulic acid is bound with arabinoxylans by ester bonds, and most of the *p*-coumaric acid molecules are also linked by ester bonds to coryngyl monolignol units of lignin present in the cell wall. During fermentation, the enzymes produced by the LAB degraded the ester bonds and caused the release of the phenolic acids of ferulic and *p*-coumaric, leading to a rise in TPC (Xie, Guo, Li, Wu, & Yu, 2021). These findings are in agreement with those declared by Spaggiari et al. (2020) who utilized solid-state fermentation to raise the TPC of wheat bran. They held endoxylanases, arabinofuranosidases, and ferulic acid esterases produced by *Lactobacillus rhamnosus* during fermentation, responsible for the increase in the TPC of the fermented wheat bran.

The highest TPC of the fermented samples was observed on the 2nd day of fermentation. The TPCs of the corn brans fermented by *Lactobacillus reuteri* (CBLR) and *Lactobacillus plantarum* (CBLP) respectively increased by 412% (3.25 mg GAE/g dwb) and 335% (2.74 mg GAE/g dwb), compared with that of the control. This revealed that *L. reuteri* was more able than *L. plantarum* to elevate the TPC during fermentation, probably due to the higher capability of *L. reuteri* in producing the enzymes, especially ferulic acid esterases. Therefore, *L. reuteri* was more effective than *L. plantarum* in releasing the esterified phenolic compounds and converting them into free ones.

It is interesting to note that the TPC started to decline after 48 h of fermentation, which can be ascribed to the decrease in the micronutrients of the culture medium during fermentation. Owing to the availability of the micronutrients in the early fermentation, the LAB could produce enzymes which were able to hydrolyze bound antioxidative compounds, thus releasing free phenolic acids. However, the content of the available micronutrients may have lowered after the 2nd day of fermentation (Călinoiu, Cătoi, & Vodnar, 2019). Furthermore, it

Table 1

Effect of fermentation time on total phenolic content, DPPH and ABTS radicals scavenging activity of the control and fermented corn bran samples.

Time (h)	TPC (mg GAE/g dwb)		DPPH scavenging activity (µmol TE/g dwb)		ABTS scavenging activity (%)	
	CBLR*	CBLP*	CBLR	CBLP	CBLR	CBLP
0	0.63 ^g ± 0.05	0.63 ^f ± 0.05	295.6 ^d ± 4.3	295.6 ^d ± 4.3	11.97 ^e ± 2.25	11.97 ^e ± 2.25
12	1.85 ^f ± 0.07	1.69 ^e ± 0.10	409.0 ^c ± 2.7	411.6 ^c ± 1.9	40.63 ^d ± 1.94	34.25 ^d ± 1.85
24	2.32 ^e ± 0.04	2.09 ^d ± 0.08	435.6 ^b ± 6.7	424.3 ^b ± 4.2	50.15 ^c ± 3.08	37.98 ^{cd} ± 2.76
36	2.74 ^d ± 0.10	2.51 ^c ± 0.06	434.5 ^b ± 1.5	426.8 ^b ± 7.1	59.88 ^b ± 2.61	43.41 ^{bc} ± 2.93
48	3.25 ^a ± 0.08	2.74 ^a ± 0.04	456.6 ^a ± 6.2	440.6 ^a ± 5.0	64.83 ^a ± 2.28	47.04 ^{ab} ± 3.25
60	3.09 ^b ± 0.06	2.65 ^b ± 0.03	446.9 ^a ± 3.2	442.9 ^a ± 7.0	65.84 ^a ± 1.35	50.63 ^a ± 1.09
72	2.95 ^c ± 0.06	2.41 ^c ± 0.06	454.1 ^a ± 5.2	439.0 ^a ± 4.5	59.16 ^b ± 3.12	47.58 ^{ab} ± 2.30

* CBLR, fermented corn bran by *Lactobacillus reuteri*; CBLP, fermented corn bran by *Lactobacillus plantarum*. Values are mean ± SD (n = 3). Different superscript lowercase letters in each column indicate significant differences ($p < 0.05$).

has been reported that some LAB such as *Lactobacillus hilgardii*, *Lactobacillus brevis*, and *Lactobacillus plantarum* are able to bioconvert phenolic compounds and can change ferulic acid into compounds like vanillin and vanillyl alcohol through non-oxidative decarboxylation (Kaur, Chakraborty, & Kumar, 2013). Li et al. (2018) extracted apple phenolic compounds using solid-state fermentation by *L. plantarum*. They declared a reduction in TPC after 72 h of fermentation and attributed this to the bioconversion of simple phenolics and the depolymerization of high-molecular-weight phenolic compounds by *L. plantarum*.

3.2. Antioxidant activity

The changes in the antioxidant activity of the fermented corn bran extracts during fermentation were measured using the DPPH and ABTS assays (Table 1). The purple color of the DPPH[•] solution and the blue-green color of the ABTS^{•+} solution disappear in the presence of antioxidant compounds. The degree of discoloration denotes the free radicals scavenging activity of the extracts (Salar, Certik, & Brezova, 2012).

During fermentation, a significant rise was observed in the antioxidant activity of the fermented samples compared with the control. The

DPPH scavenging activity increased from 295.6 $\mu\text{mol TE/g dwb}$ for the control to 456.6 $\mu\text{mol TE/g dwb}$ for CBLR and 442.9 $\mu\text{mol TE/g dwb}$ for CBLP and the ABTS scavenging activity was also raised from 12.0% for the control to 65.8% for CBLR and 50.6% for CBLP, maximally.

The increase in the antioxidant activity of fermented corn brans can be due to the rise in the release of antioxidant compounds caused by microbial hydrolysis. The enzymes produced by the LAB brought about the release of various antioxidants by destructing the cell wall. In addition, during fermentation, microorganisms are exposed to oxidative stress, which may lead to the evolution of protective mechanisms involving enzymatic antioxidation. This mechanism may contribute to the antioxidative effect of fermentation (Calinoiu et al., 2019). Moreover, the proteases and peptidases produced by the LAB during fermentation might have caused the release of peptides with antioxidant activity (Pontonio, Dingeo, Gobbetti, & Rizzello, 2019). These results are consistent with those reported by Sawangwan, Porncharoenp, and Nimraksa (2021) who maintained that the fermentation of rice bran by LAB increased the antioxidant activity by 45–55%.

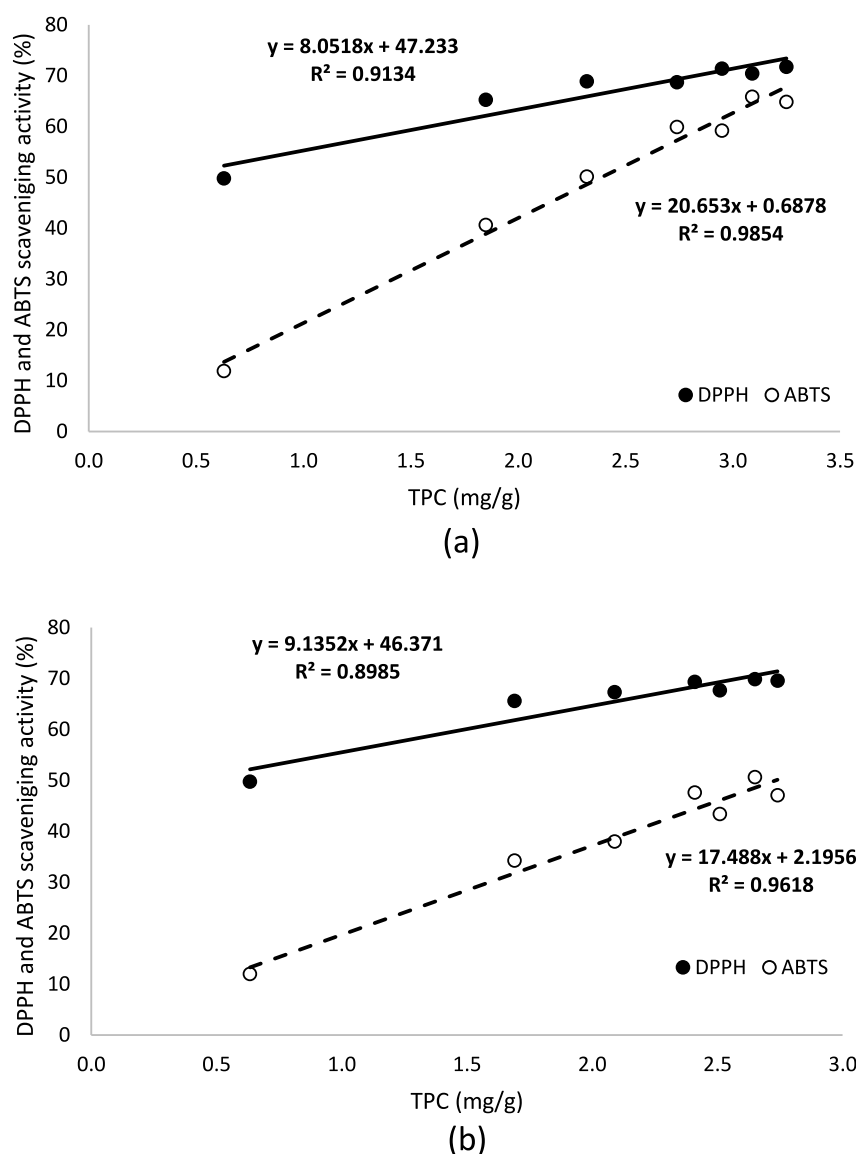


Fig. 1. Correlation between TPC and antioxidant activity of ethanolic extract of corn bran samples fermented by a) *Lactobacillus reuteri* b) *Lactobacillus plantarum*.

3.3. Correlation between TPC and antioxidant activity of ethanolic extracts

The determination coefficient was computed by correlating the antioxidant activity and TPC of the fermented samples. A strong correlation was observed between the TPC and antioxidant activity of the ethanolic extracts of the corn bran during fermentation (Fig. 1). This demonstrated that the elevated antioxidant activity of the fermented samples could be caused by the increased TPC during fermentation (Salar et al., 2012), although the antioxidant activity of phenolic compounds depends on their chemical structure, as well as the number and orientation of the hydroxyl groups bound with their aromatic rings (Punia et al., 2020). For instance, it was claimed that the presence of (CH)₂-COOH group was the main reason behind the antioxidant activity of most hydroxycinnamic acids compared with hydroxybenzoic acids (Calinoiu et al., 2019).

The determination coefficient (R^2) between TPC and DPPH scavenging activity was found to be 0.91 for CBLR and 0.89 for CBLP. Similarly, the coefficient between TPC and ABTS scavenging activity was equal to 0.98 for CBLR and 0.96 for CBLP. In several studies, significant positive correlations have been observed between the TPC and antioxidant activity of corn and the antioxidant extracts of other plants. Sandhu, Punia, and Kaur (2016) reported a positive correlation between the TPC and DPPH scavenging activity of the wheat fermented by *Aspergillus awamori*.

3.4. ORAC

This assay directly reveals the antioxidant activity associated with the breakdown of peroxy radical chains by monitoring the inhibition of peroxy-free radicals (Preedy & Patel, 2021). Fig. 2 illustrates the fluorescence decay profiles of the fermented and control samples as well as Trolox 20 µmol/L. The AUCs of CBLR (46.6 RFU.min) and CBLP (45.2 RFU.min) were significantly larger than that of the control (7.4 RFU.min). This demonstrated the higher ORAC of the fermented samples compared with the control. Similarly, Jung et al. (2017) mentioned that fermentation brought about an increase in the antioxidant activity of rice bran, measured by the ORAC assay.

3.5. Composition of phenolic acids

According to the literature, the external standards of ferulic, *p*-coumaric, syringic, caffeic, and vanillic acids were employed to quantify the phenolic acids of the fermented and control corn bran samples. The

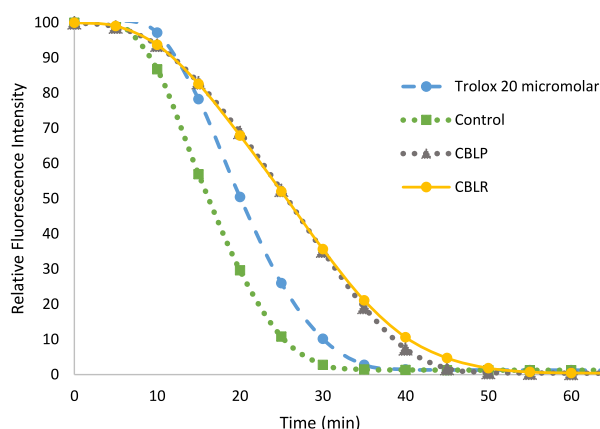


Fig. 2. Fluorescence intensity decay curve related to the control corn bran, trolox (20 µmol/L), and corn bran samples fermented by *Lactobacillus reuteri* (CBLR) and *Lactobacillus plantarum* (CBLP).

concentration of the phenolic acids was 48.1 µg/g dry matter which significantly rose during fermentation (Table 2). During solid-state fermentation, LAB can convert complex bioactive compounds into their simple free forms by producing specific enzymes, leading to an increase in the phenolic acids concentration of the fermented samples (Bangar et al., 2022).

Ferulic acid was the major phenolic compound in both fermented samples. The highest ferulic acid content was measured to be 415.2 and 372.7 µg/g dry matter on the 2nd day of fermentation for CBLR and CBLP, respectively. *p*-coumaric acid was the second major phenolic substance with 96.4 and 51.2 µg/g dry matter for CBLR and CBLP, respectively. Syringic, caffeic, and vanillic acids were the other detected phenolic acids. These findings conform to those of Wang et al. (2014) who realized that ferulic and *p*-coumaric acids were the most abundant bound phenolic acids in corn bran, and the presence and concentration of other phenolic acids varied depending on the corn variety and the extraction method. In both fermented samples, ferulic acid content declined after the 2nd day, while vanillic acid content increased during the entire fermentation. This showed that a fraction of ferulic acid probably changed into vanillic acid through bioconversion. This mechanism involves the transformation of ferulic acid to vanillin which exhibits toxic effects on microbes and can reduce to less toxic intermediates such as vanillic acid (Kaur et al., 2013). Moreover, it has been suggested that ferulic acid may either decompose or bound back to the insoluble polysaccharides on the 3rd day of fermentation. These results agree with those reported by Shin, Kim, Lee, and Lim (2019) who investigated the impact of solid-state fermentation on the phenolic acid composition of rice bran. They reported that ferulic acid content substantially decreased on the fourth day of fermentation.

The results also indicated that CBLR had higher contents of ferulic and *p*-coumaric acids than CBLP, consistent with the results of TPC. The increased content of phenolic acids in fermented samples depends on the substrate type, microbial strain used for fermentation, and the fermentation conditions such as temperature, pH, and time (Wang et al., 2014; Bangar et al., 2022).

3.6. FT-IR

The variations in the chemical bonds and functional groups of the corn bran samples after 48 h of fermentation are summarized in Fig. 3 and Table 3. Corn bran is basically composed of cellulose, hemicellulose, protein, and lignin. The absorbance peaks at 1743, 1650, and 1382 cm⁻¹

Table 2
Phenolic acid composition of fermented corn bran samples during fermentation.

Sample	Ferulic acid (µg/g)	<i>p</i> -coumaric acid (µg/g)	Caffeic acid (µg/g)	Vanillic acid (µg/g)	Syringic acid (µg/g)
Corn Bran fermented by <i>Lactobacillus reuteri</i>					
Control	35.92 ^d ± 3.26	8.00 ^d ± 1.05	0.18 ^c ± 0.05	1.04 ^d ± 0.05	2.98 ^d ± 0.09
CBLR-24 h	285.3 ^c ± 5.40	55.82 ^c ± 2.52	0.95 ^b ± 0.05	5.46 ^c ± 0.18	12.17 ^c ± 1.08
CBLR-48 h	415.2 ^a ± 6.82	96.45 ^a ± 3.65	1.23 ^a ± 0.08	8.56 ^b ± 0.46	19.97 ^a ± 1.36
CBLR-72 h	365.4 ^b ± 6.09	68.71 ^b ± 2.78	1.22 ^a ± 0.04	11.54 ^a ± 0.32	17.12 ^b ± 1.12
Corn Bran fermented by <i>Lactobacillus plantarum</i>					
Control	35.92 ^d ± 3.26	8.00 ^d ± 1.05	0.18 ^c ± 0.05	1.04 ^d ± 0.05	2.98 ^d ± 0.09
CBLP-24 h	245.8 ^c ± 4.73	31.53 ^c ± 1.95	0.74 ^b ± 0.02	5.35 ^c ± 0.20	10.43 ^b ± 1.20
CBLP-48 h	372.7 ^a ± 8.39	51.24 ^a ± 2.23	1.08 ^a ± 0.02	8.26 ^b ± 1.38	16.78 ^a ± 0.93
CBLP-72 h	282.0 ^b ± 5.56	43.32 ^b ± 2.50	1.05 ^a ± 0.04	10.89 ^a ± 0.95	15.43 ^a ± 1.82

Values are mean ± SD (n = 3). Different superscript lowercase letters in each column indicate significant differences ($p < 0.05$).

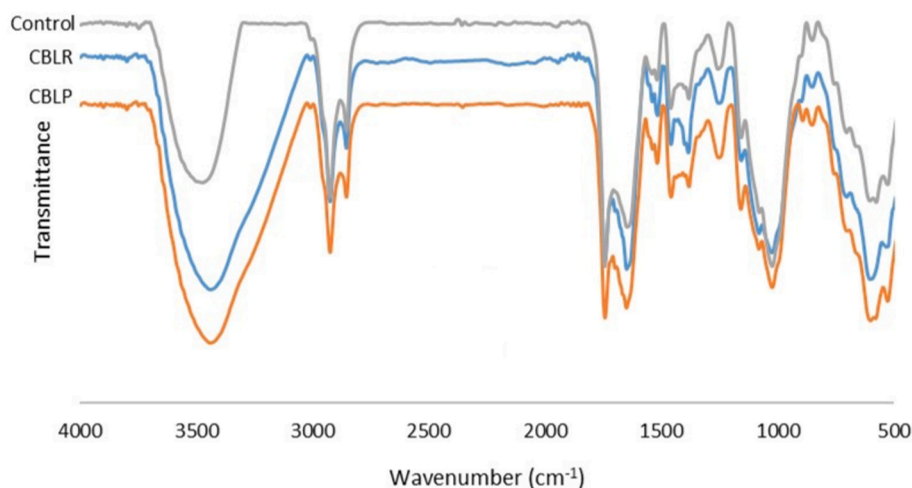


Fig. 3. FT-IR spectra of the control corn bran and fermented corn bran samples by *Lactobacillus reuteri* (CBLR) and *Lactobacillus plantarum* (CBLP).

Table 3

Assignment of principal descriptive IR absorption bands in the unfermented and fermented corn bran.

Wavenumber (cm ⁻¹)	Assignment	Characterisation	Reference
852	Aromatic CH out of plane	Lignin	Artz et al. (2008)
1025	C-O stretching	Glycosidic linkage between cellulose and hemicellulose sugar units	Pasha et al. (2021)
1157	C-O-C stretching	Polysaccharides	Waghmare, Kadam, Saratale, and Govindwar (2014)
1382	OH plane bending	Cellulose	Pasha et al. (2021)
1459	symmetric CH ₂ bending	Polysaccharides	Waghmare et al. (2014)
1519	carbonyl stretching	Aromatic ring of Lignin	Waghmare et al. (2014)
1650	carbonyl stretching	acetyl groups in hemicelluloses and for aldehydic group present in lignin	Waghmare et al. (2014)
1743	carbonyl stretching	Ester groups in hemicellulose	Pereira et al., (2019)
2925	C-H stretching	Polysaccharides and proteins	Liu et al. (2017); Waghmare et al. (2014)
3474, 3466, 3441	OH stretching	Polysaccharides and phenolic compounds	Tsui and Yang (2021)

represent hemicellulose, and the ones at 3474, 2925, 1382, and 1157 cm⁻¹ indicate cellulose. The low-intensity absorbance band at 852 cm⁻¹ denotes the low content of lignin in the samples (Liu et al., 2017).

According to the FT-IR spectra, apparent changes were observed in the functional groups of the substrate during fermentation. The most marked change in the fermented samples, compared with the control, was associated with the rise in the absorbance intensity in the wavenumber range of 3200–3500 cm⁻¹. A broad band in the range of 3200–3500 cm⁻¹, related to the OH stretching vibrations of polysaccharides and polyphenols (Tsui & Yang, 2021) shifted from 3474 cm⁻¹ in the control to 3441 and 3466 cm⁻¹ in CBLR and CBLP, respectively. Furthermore, the band was intensified in the fermented samples. This could be due to the increased content of free phenolic acids resulting from the activity of the enzymes produced by the LAB during fermentation. In addition, a variety of OH vibrations may have appeared in this region because of the formation of strong hydrogen

bonds between the phenolic acids, leading to the formation of phenolic acid dimers.

The other change occurred in the wavenumber range of 2800–3000 cm⁻¹ with a peak absorbance at 2925 cm⁻¹, indicating the stretching vibrations of CH₂ and CH₃ (alkyls). Fermentation lowered the absorbance intensity of the corn bran in this region. Moreover, the absorbance at 1743 cm⁻¹, revealing ester bonds, was less intense for the fermented samples than for the control, probably because the LAB played a key role in the degradation of the ester bonds of hemicellulose and lignin and released free phenolic acids by producing esterases during fermentation (Xie et al., 2021).

The intensity of the absorbance at 1025 cm⁻¹ was lower for the fermented samples than for the control. Endoxylanases including the cellulases and xylanases produced during fermentation, as the enzymes decomposing the main chain of the carbohydrates of cell walls, can lower the absorbance intensity at this wavenumber by cleaving the intramolecular glycosidic bonds (β-1,4) of the hemicellulose chains. These findings conform to those of Pasha, Ahmad, and Usman (2021) who declared that native wheat bran had a higher absorbance intensity at 1030 cm⁻¹ than the enzymatically-treated samples, owing to possessing intact oligosaccharides.

3.7. SEM

SEM was applied to examine morphological alterations in the fermented and control corn samples. The SEM images (Fig. 4) depicted that the control had a coarse structure with polyhedral particles and sharp boundaries. In contrast, solid-state fermentation altered the surface of the corn bran. Obviously, the surface of the fermented samples had a more melted, more loosened, more deformed, and smoother structure containing agglomerated particles. Additionally, the surface of the particles was scratched in the fermented samples; thus, the boundaries between their particles cannot be easily seen. These changes may be attributed to the exopolysaccharide enzymes produced by the LAB during fermentation.

The more pronounced presence of pores in some parts of the fermented samples demonstrates the enzymatic activity of the LAB, confirming the more antioxidant activity of the fermented samples compared with the control. These observations are in agreement with those of Akpabli-Tsigbe et al. (2021) who cited that brown rice starch granules lost their regular polyhedral shapes because of the activity of different enzymes during fermentation.

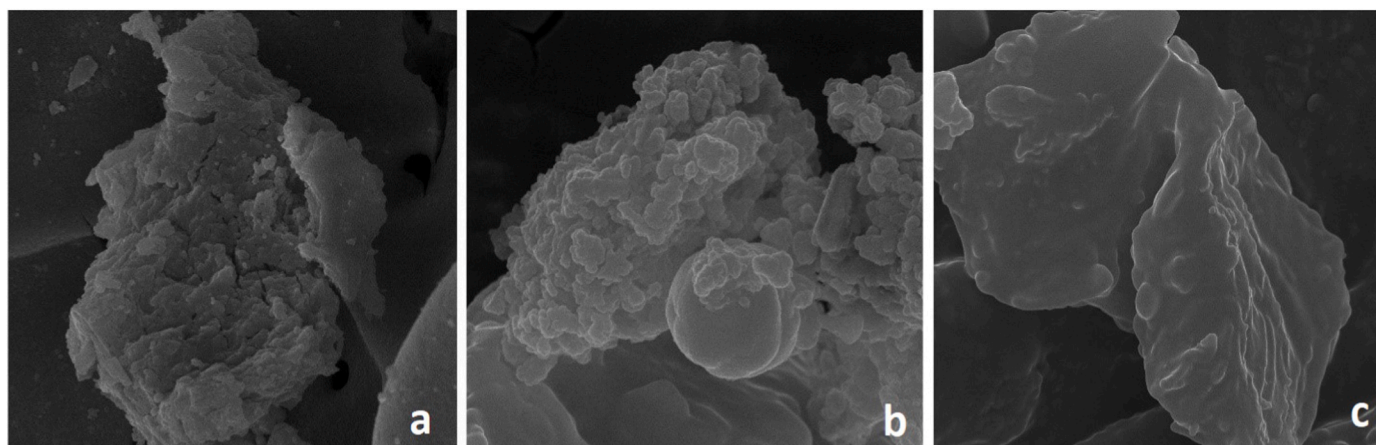


Fig. 4. Scanning electron microscopy images of corn bran samples a) control b) fermented by *Lactobacillus reuteri* c) fermented by *Lactobacillus plantarum*.

4. Conclusions

This study demonstrates that the solid-state fermentation of corn bran with *L. reuteri* and *L. plantarum* significantly increased the results of TPC, DPPH, ABTS and ORAC assays. The results showed that the fermented corn bran had greater antioxidant activity, compared with the unfermented sample. The optimal fermentation duration was equal to two days in terms of TPC and antioxidant activity. Five standards of ferulic, *p*-coumaric, caffeic, vanillic, and syringic acids were chosen to screen bioactive compounds using HPLC. The solid-state fermentation increased the phenolic acid content in the alcoholic extracts of the fermented corn bran samples. The obtained outcome has shown that fermentation is an effective method for enhancing bioactive metabolites in corn bran and can change this substrate into a good candidate for the formulation of functional foods and nutraceuticals in food industries. More research is needed to optimize the size and age of inoculum as well as extraction parameters to standardize the fermentation process.

CRediT authorship contribution statement

Mehdi Akbari: Writing – original draft, preparation, Investigation, Methodology, Software, Validation, Formal analysis. **Seyed Hadi Razavi:** Conceptualization, Supervision, Writing – review & editing. **Faramarz Khodaiyan:** Conceptualization, Supervision, Writing – review & editing. **Jesús Blesa:** Supervision, Writing – review & editing, Methodology. **María J. Esteve:** Supervision, Writing – review & editing, Resources, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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